Time Series Analysis - Practical 10

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Practical 10

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Question

Consider the "AirPassengers" data from the R library and write the R program to forecast the monthly airline passenger numbers for the next 12 months.

Understanding the Question

This practical involves forecasting monthly airline passenger numbers using the classic AirPassengers dataset. The dataset contains monthly totals of international airline passengers from 1949 to 1960.

Key Objectives: - Analyze the pattern and seasonality in airline passenger data - Decompose the time series into trend, seasonal, and random components - Check for stationarity and apply transformations if needed - Compare double and triple exponential smoothing methods - Generate 12-month ahead forecasts - Validate model performance and draw conclusions

Methodology

Step-by-Step Approach:

1. **Data Exploration**
   - Load the AirPassengers dataset
   - Visualize the original series
   - Identify trend and seasonal patterns
2. **Time Series Decomposition**
   - Apply multiplicative decomposition
   - Extract and analyze trend, seasonal, and random components
3. **Stationarity Testing**
   - Plot ACF and PACF
   - Perform KPSS test for stationarity
   - Apply log transformation to stabilize variance
4. **Model Identification**

- Fit double exponential smoothing (Holt's method) - without seasonality
- Fit triple exponential smoothing (Holt-Winters) - with seasonality
- Compare SSE (Sum of Squared Errors)

5. **Model Estimation**
   - Estimate smoothing parameters (alpha, beta, gamma)
   - Analyze seasonal indices
6. **Forecasting**
   - Generate 12-month forecasts using both models
   - Compare forecast performance
   - Plot forecasts with confidence intervals

R Code and Analysis

```r
# Clear workspace
rm(list = ls(all = TRUE))

# Load required libraries
library(tseries)
library(forecast)
library(TTR)
library(lmtest)

# Step 1: Data Calling
data("AirPassengers")

cat("AirPassengers Dataset:\n")
```

```
## AirPassengers Dataset:
```

```r
print(AirPassengers)
```

```
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
```

```r
cat("\nDataset Summary:\n")
```

```
##
## Dataset Summary:
```

```r
print(summary(AirPassengers))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   104.0   180.0   265.5   280.3   360.5   622.0
```

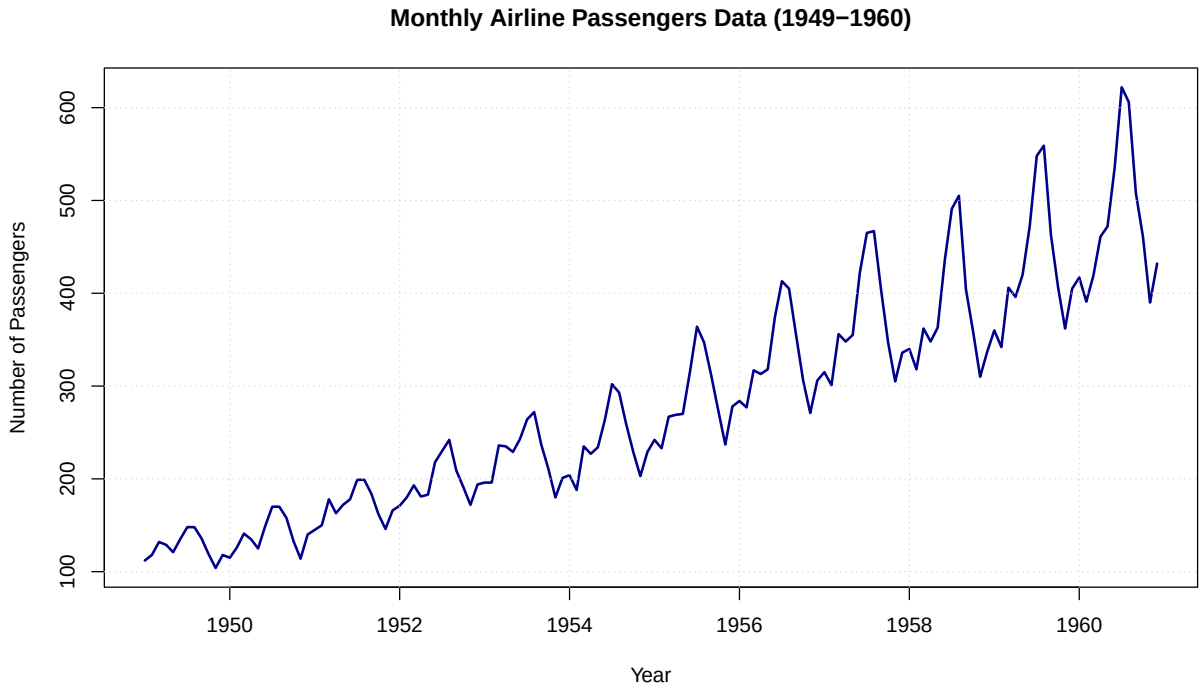
```r
cat("\nDataset Structure:\n")
```

```
##
## Dataset Structure:
```

```r
print(str(AirPassengers))
```

```
##  Time-Series [1:144] from 1949 to 1961: 112 118 132 129 121 135 148 148 136 119 ...
## NULL
```

```r
# Plot original data
plot(AirPassengers,
     main = "Monthly Airline Passengers Data (1949-1960)",
     xlab = "Year",
     ylab = "Number of Passengers",
     lwd = 2,
     col = "darkblue",
     cex.main = 1.2,
     cex.lab = 1.1)
grid(col = "lightgray", lty = "dotted")
```



```r
# Step 2: Time Series Decomposition
AirPassengers_decompose <- decompose(AirPassengers, type = "multiplicative")

cat("\nDecomposition Components:\n")
```

```
##
## Decomposition Components:
```

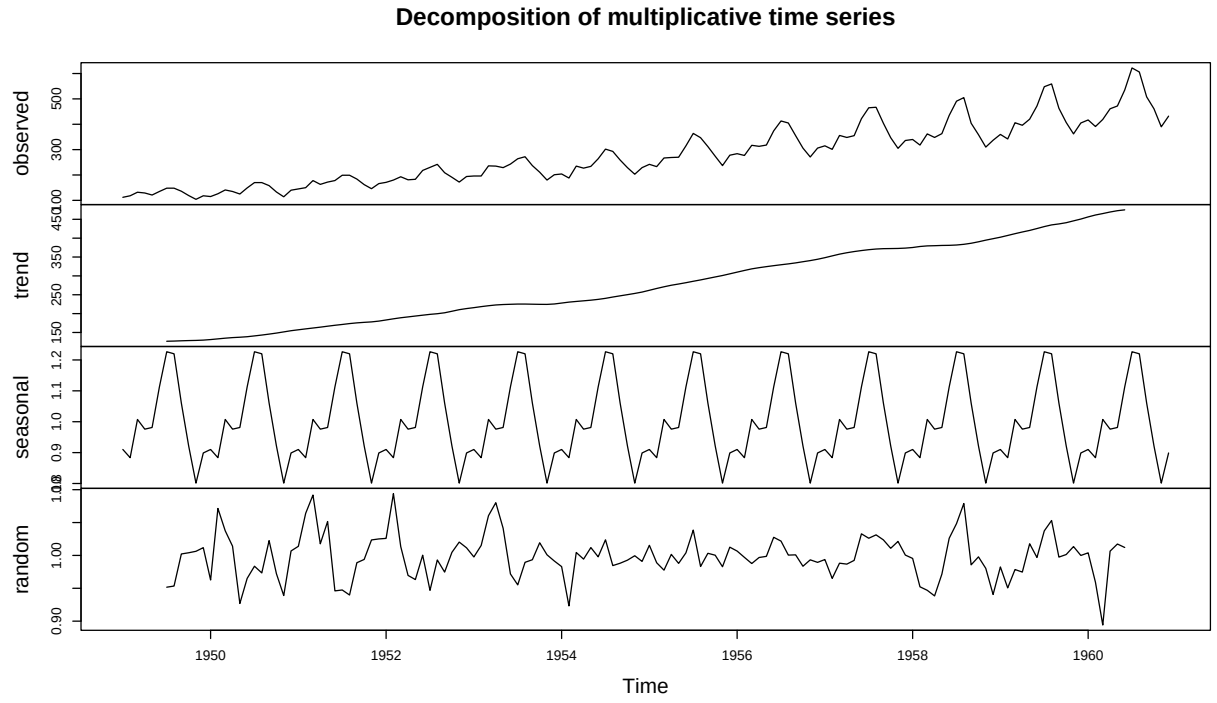
```r
print(AirPassengers_decompose)
```

```
## $x
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
##
## $seasonal
##             Jan       Feb       Mar       Apr       May       Jun       Jul
## 1949 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1950 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1951 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1952 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
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## 1955 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1956 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1957 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
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## 1959 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
## 1960 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
##             Aug       Sep       Oct       Nov       Dec
## 1949 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1950 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1951 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1952 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1953 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1954 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1955 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1956 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1957 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1958 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1959 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
## 1960 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
```

```
##
## $trend
##           Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
## 1949       NA       NA       NA       NA       NA       NA 126.7917 127.2500
## 1950 131.2500 133.0833 134.9167 136.4167 137.4167 138.7500 140.9167 143.1667
## 1951 157.1250 159.5417 161.8333 164.1250 166.6667 169.0833 171.2500 173.5833
## 1952 183.1250 186.2083 189.0417 191.2917 193.5833 195.8333 198.0417 199.7500
## 1953 215.8333 218.5000 220.9167 222.9167 224.0833 224.7083 225.3333 225.3333
## 1954 228.0000 230.4583 232.2500 233.9167 235.6250 237.7500 240.5000 243.9583
## 1955 261.8333 266.6667 271.1250 275.2083 278.5000 281.9583 285.7500 289.3333
## 1956 309.9583 314.4167 318.6250 321.7500 324.5000 327.0833 329.5417 331.8333
## 1957 348.2500 353.0000 357.6250 361.3750 364.5000 367.1667 369.4583 371.2083
## 1958 375.2500 377.9167 379.5000 380.0000 380.7083 380.9583 381.8333 383.6667
## 1959 402.5417 407.1667 411.8750 416.3333 420.5000 425.5000 430.7083 435.1250
## 1960 456.3333 461.3750 465.2083 469.3333 472.7500 475.0417       NA       NA
##           Sep      Oct      Nov      Dec
## 1949 127.9583 128.5833 129.0000 129.7500
## 1950 145.7083 148.4167 151.5417 154.7083
## 1951 175.4583 176.8333 178.0417 180.1667
## 1952 202.2083 206.2500 210.4167 213.3750
## 1953 224.9583 224.5833 224.4583 225.5417
## 1954 247.1667 250.2500 253.5000 257.1250
## 1955 293.2500 297.1667 301.0000 305.4583
## 1956 334.4583 337.5417 340.5417 344.0833
## 1957 372.1667 372.4167 372.7500 373.6250
## 1958 386.5000 390.3333 394.7083 398.6250
## 1959 437.7083 440.9583 445.8333 450.6250
## 1960       NA       NA       NA       NA
##
## $random
##           Jan       Feb       Mar       Apr       May       Jun       Jul
## 1949       NA        NA        NA        NA        NA        NA 0.9516643
## 1950 0.9626030 1.0714668 1.0374474 1.0140476 0.9269030 0.9650406 0.9835566
## 1951 1.0138446 1.0640180 1.0918541 1.0176651 1.0515825 0.9460444 0.9474041
## 1952 1.0258814 1.0939696 1.0134734 0.9695596 0.9632673 1.0003735 0.9468562
## 1953 0.9976684 1.0151646 1.0604644 1.0802327 1.0413329 0.9718056 0.9551933
## 1954 0.9829785 0.9232032 1.0044417 0.9943899 1.0119479 0.9978740 1.0237753
## 1955 1.0154046 0.9888241 0.9775844 1.0015732 0.9878755 1.0039635 1.0385512
## 1956 1.0066157 0.9970250 0.9876248 0.9968224 0.9985644 1.0275560 1.0217685
## 1957 0.9937293 0.9649918 0.9881769 0.9867637 0.9924177 1.0328601 1.0261250
## 1958 0.9954212 0.9522762 0.9469115 0.9383993 0.9715785 1.0261340 1.0483841
## 1959 0.9825176 0.9505736 0.9785278 0.9746440 1.0177637 0.9968613 1.0373136
## 1960 1.0039279 0.9590794 0.8940857 1.0064948 1.0173588 1.0120790        NA
##           Aug       Sep       Oct       Nov       Dec
## 1949 0.9534014 1.0022198 1.0040278 1.0062701 1.0118119
## 1950 0.9733720 1.0225047 0.9721928 0.9389527 1.0067914
## 1951 0.9397599 0.9888637 0.9938809 1.0235337 1.0250824
## 1952 0.9931171 0.9746302 1.0046687 1.0202797 1.0115407
## 1953 0.9894989 0.9934337 1.0192680 1.0009392 0.9915039
## 1954 0.9845184 0.9881036 0.9927613 0.9995143 0.9908692
## 1955 0.9831117 1.0032501 1.0003084 0.9827720 1.0125535
## 1956 1.0004765 1.0008730 0.9835071 0.9932761 0.9894251
## 1957 1.0312668 1.0236147 1.0108432 1.0212995 1.0005263
## 1958 1.0789695 0.9856540 0.9977971 0.9802940 0.9405687
```

```
## 1959 1.0531001 0.9974447 1.0013371 1.0134608 0.9999192
## 1960        NA        NA        NA        NA        NA
##
## $figure
##  [1] 0.9102304 0.8836253 1.0073663 0.9759060 0.9813780 1.1127758 1.2265555
##  [8] 1.2199110 1.0604919 0.9217572 0.8011781 0.8988244
##
## $type
## [1] "multiplicative"
##
## attr(,"class")
## [1] "decomposed.ts"
```

```r
# Plot decomposition
plot(AirPassengers_decompose)
```




```r
# Extract seasonal indices
seasonal_indices <- AirPassengers_decompose$figure
names(seasonal_indices) <- month.abb
cat("\nSeasonal Indices:\n")
```

```
##
## Seasonal Indices:
```

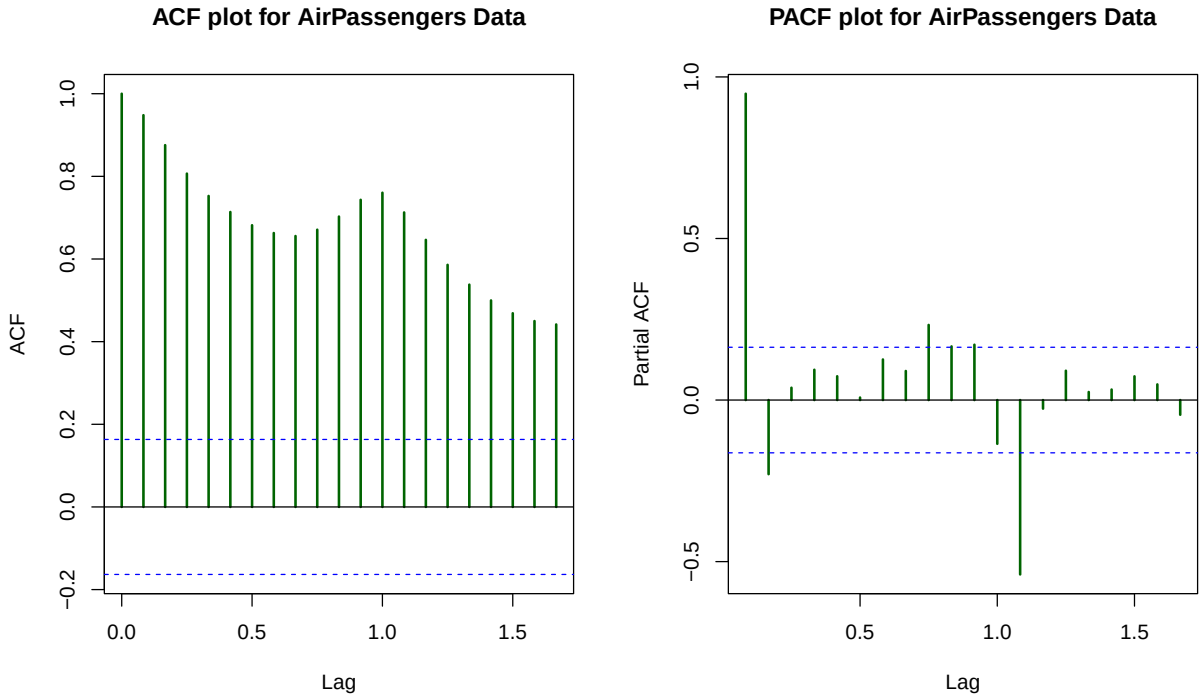
```r
print(round(seasonal_indices, 4))
```

```
##     Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep     Oct     Nov
## 0.9102 0.8836 1.0074 0.9759 0.9814 1.1128 1.2266 1.2199 1.0605 0.9218 0.8012
```

```
##    Dec
## 0.8988
```

```r
# Step 3: Checking Stationarity and Non-stationarity
par(mfrow = c(1, 2))
acf(AirPassengers,
    main = "ACF plot for AirPassengers Data",
    lag.max = 20,
    lwd = 2,
    col = "darkgreen")

pacf(AirPassengers,
     main = "PACF plot for AirPassengers Data",
     lag.max = 20,
     lwd = 2,
     col = "darkgreen")
```




```r
par(mfrow = c(1, 1))

cat("\nKPSS Test for Stationarity:\n")
```

```
##
## KPSS Test for Stationarity:
```

```r
kpss_result <- kpss.test(AirPassengers)
print(kpss_result)
```

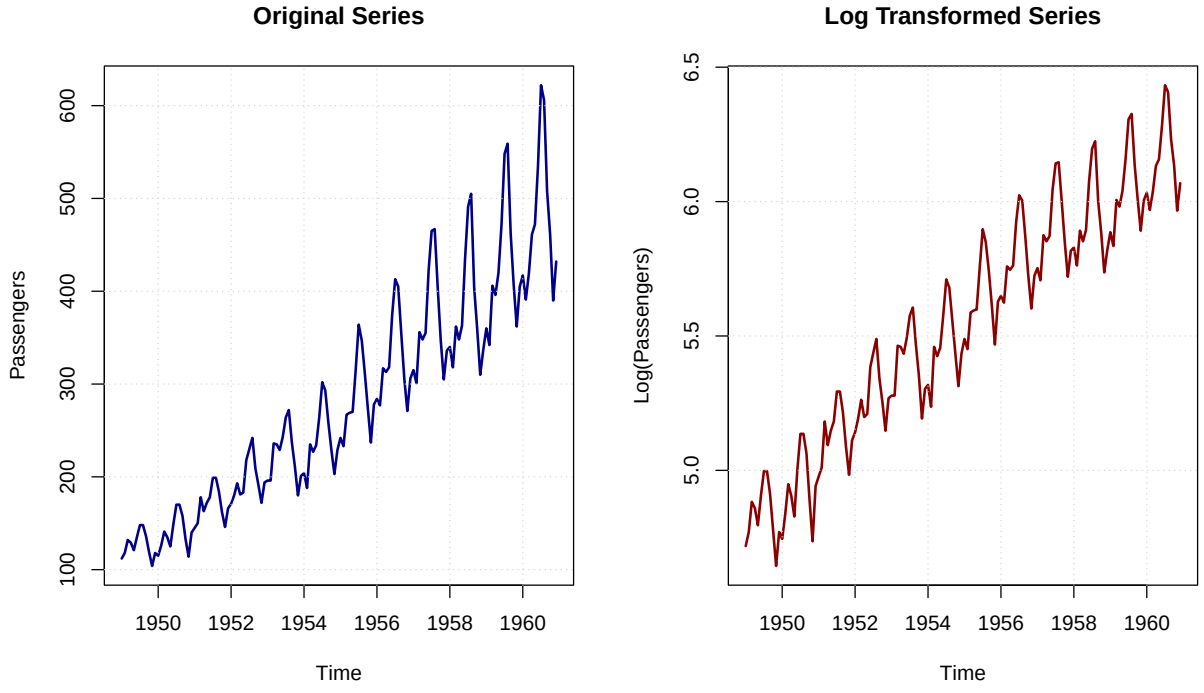
```
##
```

```
##  KPSS Test for Level Stationarity
##
## data:  AirPassengers
## KPSS Level = 2.7395, Truncation lag parameter = 4, p-value = 0.01
```

```r
# Step 4: Using Transformation
Air_Log <- log(AirPassengers)

par(mfrow = c(1, 2))
plot(AirPassengers,
     main = "Original Series",
     ylab = "Passengers",
     col = "darkblue",
     lwd = 2)
grid(col = "lightgray", lty = "dotted")

plot(Air_Log,
     main = "Log Transformed Series",
     ylab = "Log(Passengers)",
     col = "darkred",
     lwd = 2)
grid(col = "lightgray", lty = "dotted")
```




```r
par(mfrow = c(1, 1))
```

```r
# Step 5: Model Identification - Double Exponential Smoothing
AirPassengers_Holtwinter2 <- HoltWinters(log(AirPassengers), gamma = FALSE)

cat("\nDouble Exponential Smoothing (Holt's Method - No Seasonality):\n")
```

```
##
## Double Exponential Smoothing (Holt's Method - No Seasonality):
```

```
print(AirPassengers_Holtwinter2)
```

```
## Holt-Winters exponential smoothing with trend and without seasonal component.
##
## Call:
## HoltWinters(x = log(AirPassengers), gamma = FALSE)
##
## Smoothing parameters:
##  alpha: 1
##  beta : 0.02443081
##  gamma: FALSE
##
## Coefficients:
##         [,1]
## a 6.06842559
## b 0.00684448
```

```
cat("\nSmoothing Parameters:\n")
```

```
##
## Smoothing Parameters:
```

```
cat("Alpha (level):", AirPassengers_Holtwinter2$alpha, "\n")
```

```
## Alpha (level): 1
```

```
cat("Beta (trend):", AirPassengers_Holtwinter2$beta, "\n")
```

```
## Beta (trend): 0.02443081
```

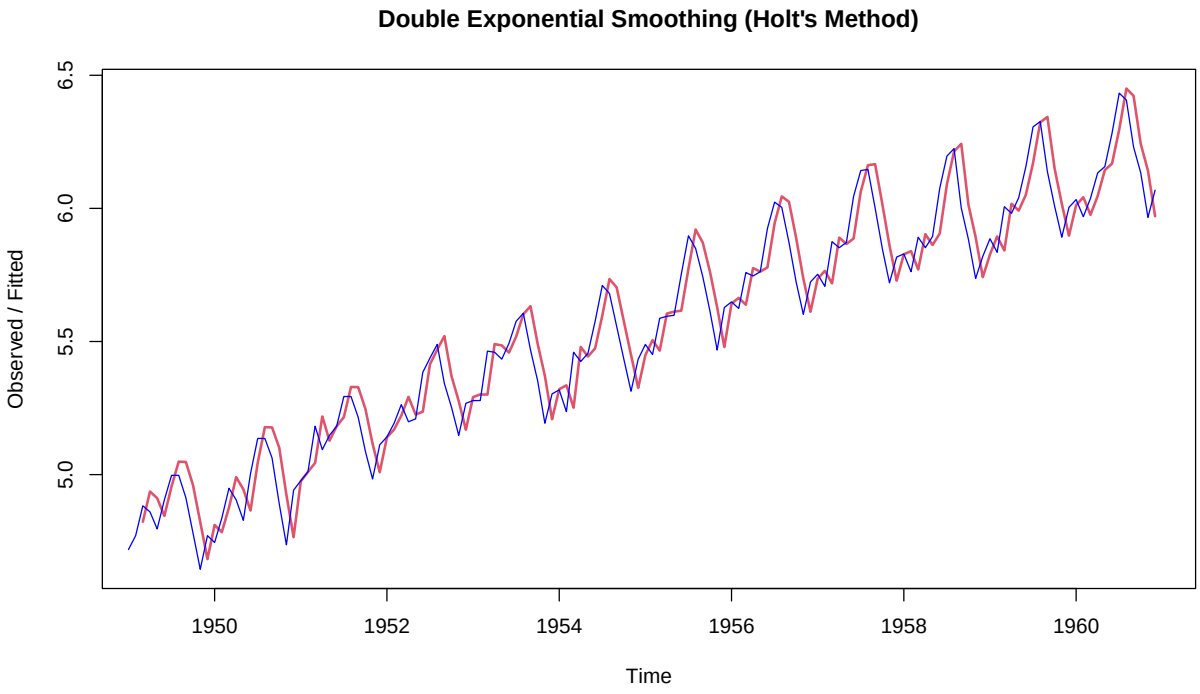
```
cat("Gamma (seasonal): FALSE\n")
```

```
## Gamma (seasonal): FALSE
```

```
cat("\nSSE (Sum of Squared Errors):", AirPassengers_Holtwinter2$SSE, "\n")
```

```
##
## SSE (Sum of Squared Errors): 1.683243
```

```
# Plot double exponential smoothing
plot(AirPassengers_Holtwinter2,
     main = "Double Exponential Smoothing (Holt's Method)",
     col = "blue",
     lwd = 2)
```




```
# Triple Exponential Smoothing (Holt-Winters with Seasonality)
AirPassengers_Holtwinter3 <- HoltWinters(log(AirPassengers))

cat("\nTriple Exponential Smoothing (Holt-Winters with Seasonality):\n")
```

```
##
## Triple Exponential Smoothing (Holt-Winters with Seasonality):
```

```
print(AirPassengers_Holtwinter3)
```

```
## Holt-Winters exponential smoothing with trend and additive seasonal component.
##
## Call:
## HoltWinters(x = log(AirPassengers))
##
## Smoothing parameters:
##  alpha: 0.3266015
##  beta : 0.005744138
##  gamma: 0.8206654
##
## Coefficients:
##             [,1]
## a    6.172308435
## b    0.008981893
## s1  -0.073201087
## s2  -0.140973564
## s3  -0.036703294
## s4   0.014522733
```

```
## s5   0.032554237
## s6   0.154873570
## s7   0.294317062
## s8   0.276063997
## s9   0.088237657
## s10 -0.032657089
## s11 -0.198012716
## s12 -0.102863837
```

```
cat("\nSmoothing Parameters:\n")
```

```
##
## Smoothing Parameters:
```

```
cat("Alpha (level):", AirPassengers_Holtwinter3$alpha, "\n")
```

```
## Alpha (level): 0.3266015
```

```
cat("Beta (trend):", AirPassengers_Holtwinter3$beta, "\n")
```

```
## Beta (trend): 0.005744138
```

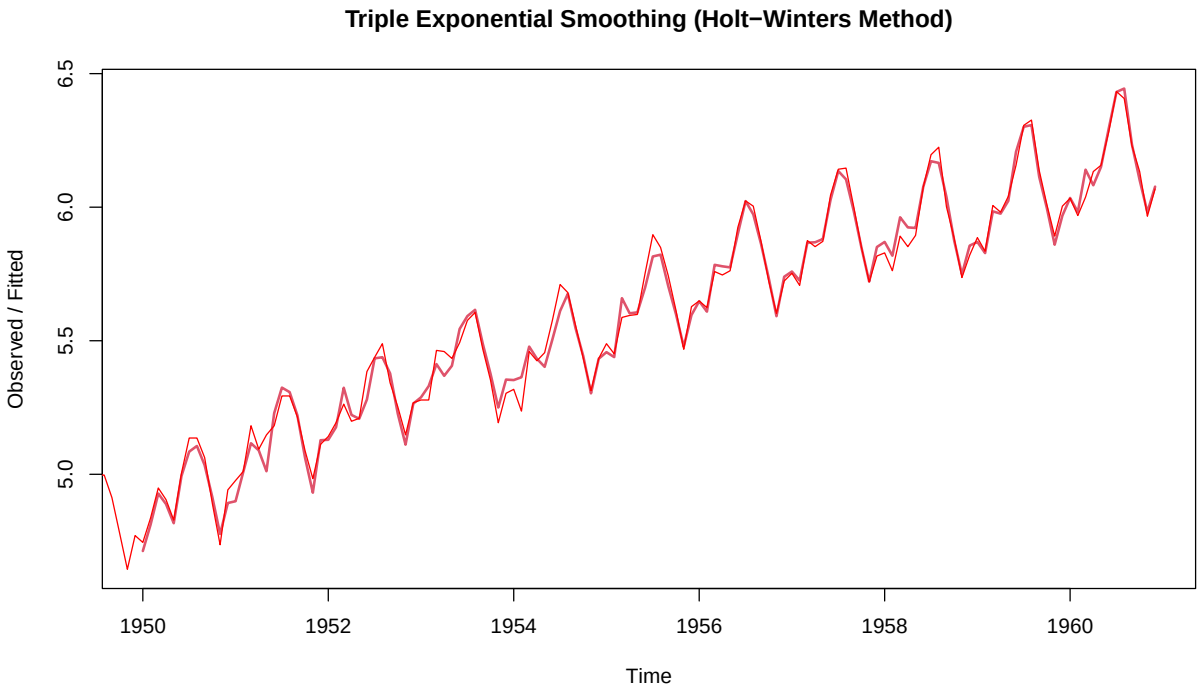
```
cat("Gamma (seasonal):", AirPassengers_Holtwinter3$gamma, "\n")
```

```
## Gamma (seasonal): 0.8206654
```

```
cat("\nSSE (Sum of Squared Errors):", AirPassengers_Holtwinter3$SSE, "\n")
```

```
##
## SSE (Sum of Squared Errors): 0.2030765
```

```
# Plot triple exponential smoothing
plot(AirPassengers_Holtwinter3,
     main = "Triple Exponential Smoothing (Holt-Winters Method)",
     col = "red",
     lwd = 2)
```




```r
# Extract seasonal coefficients
seasonal_coef <- tail(coef(AirPassengers_Holtwinter3), 12)
names(seasonal_coef) <- month.abb
cat("\nSeasonal Coefficients (Additive in Log Scale):\n")
```

```
##
## Seasonal Coefficients (Additive in Log Scale):
```

```r
print(round(seasonal_coef, 4))
```

```
##     Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep     Oct
## -0.0732 -0.1410 -0.0367  0.0145  0.0326  0.1549  0.2943  0.2761  0.0882 -0.0327
##     Nov     Dec
## -0.1980 -0.1029
```

```r
# Step 6: Model Comparison - FIXED (removed AIC)
comparison_df <- data.frame(
  Model = c("Double Exponential (No Seasonality)", "Triple Exponential (With Seasonality)"),
  SSE = round(c(AirPassengers_Holtwinter2$SSE, AirPassengers_Holtwinter3$SSE), 4),
  Parameters = c("Alpha, Beta", "Alpha, Beta, Gamma"),
  `Level (Alpha)` = c(round(AirPassengers_Holtwinter2$alpha, 4),
                      round(AirPassengers_Holtwinter3$alpha, 4)),
  `Trend (Beta)` = c(round(AirPassengers_Holtwinter2$beta, 4),
                     round(AirPassengers_Holtwinter3$beta, 4)),
  `Seasonal (Gamma)` = c("FALSE", round(AirPassengers_Holtwinter3$gamma, 4)),
  check.names = FALSE
)

cat("\nModel Comparison:\n")
```

```
##
## Model Comparison:
```

```
print(comparison_df)
```

```
##                                      Model    SSE         Parameters
##         Double Exponential (No Seasonality) 1.6832        Alpha, Beta
## gamma Triple Exponential (With Seasonality) 0.2031 Alpha, Beta, Gamma
##       Level (Alpha) Trend (Beta) Seasonal (Gamma)
##              1.0000       0.0244            FALSE
## gamma        0.3266       0.0057           0.8207
```

```
# Calculate improvement
improvement <- (AirPassengers_Holtwinter2$SSE - AirPassengers_Holtwinter3$SSE) /
               AirPassengers_Holtwinter2$SSE * 100
cat("\nSSE Improvement with Seasonality:", round(improvement, 2), "%\n")
```

```
##
## SSE Improvement with Seasonality: 87.94 %
```

```
# Step 7: Forecasting using Triple Exponential Smoothing
ForecastFinal <- forecast(AirPassengers_Holtwinter3, h = 12)

cat("\n12-Month Forecast using Triple Exponential Smoothing:\n")
```

```
##
## 12-Month Forecast using Triple Exponential Smoothing:
```

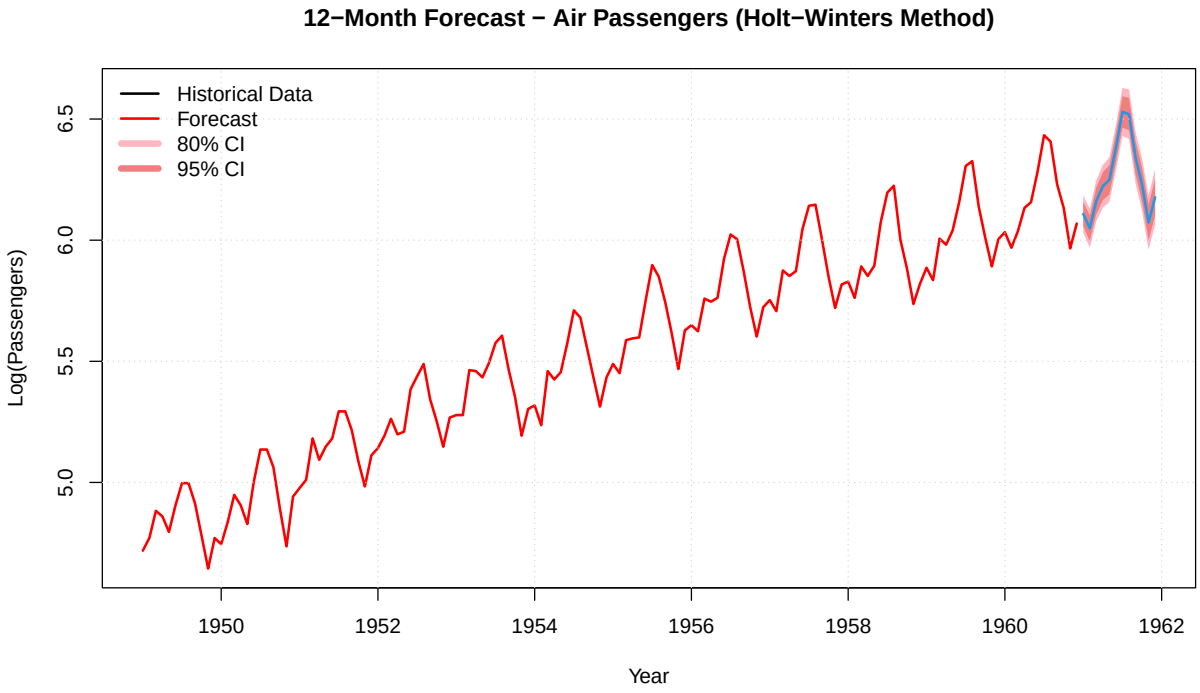
```
print(ForecastFinal)
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 1961       6.108089 6.057867 6.158312 6.031281 6.184898
## Feb 1961       6.049299 5.996436 6.102161 5.968452 6.130145
## Mar 1961       6.162551 6.107146 6.217956 6.077816 6.247285
## Apr 1961       6.222759 6.164896 6.280622 6.134265 6.311252
## May 1961       6.249772 6.189525 6.310019 6.157633 6.341912
## Jun 1961       6.381073 6.318508 6.443638 6.285388 6.476758
## Jul 1961       6.529499 6.464674 6.594324 6.430357 6.628640
## Aug 1961       6.520228 6.453195 6.587261 6.417709 6.622746
## Sep 1961       6.341383 6.272189 6.410577 6.235560 6.447206
## Oct 1961       6.229470 6.158159 6.300782 6.120408 6.338532
## Nov 1961       6.073097 5.999706 6.146487 5.960855 6.185338
## Dec 1961       6.177227 6.101793 6.252662 6.061861 6.292594
```

```
# Plot final forecast
plot(ForecastFinal,
     main = "12-Month Forecast - Air Passengers (Holt-Winters Method)",
     xlab = "Year",
     ylab = "Log(Passengers)",
     lwd = 2,
     col = "red",
```

```
     flwd = 2,
     shadecols = c("lightpink", "lightcoral"),
     cex.main = 1.2,
     cex.lab = 1.1)
grid(col = "lightgray", lty = "dotted")

# Add legend
legend("topleft",
       legend = c("Historical Data", "Forecast", "80% CI", "95% CI"),
       col = c("black", "red", "lightpink", "lightcoral"),
       lty = c(1, 1, 1, 1),
       lwd = c(2, 2, 5, 5),
       bty = "n")
```

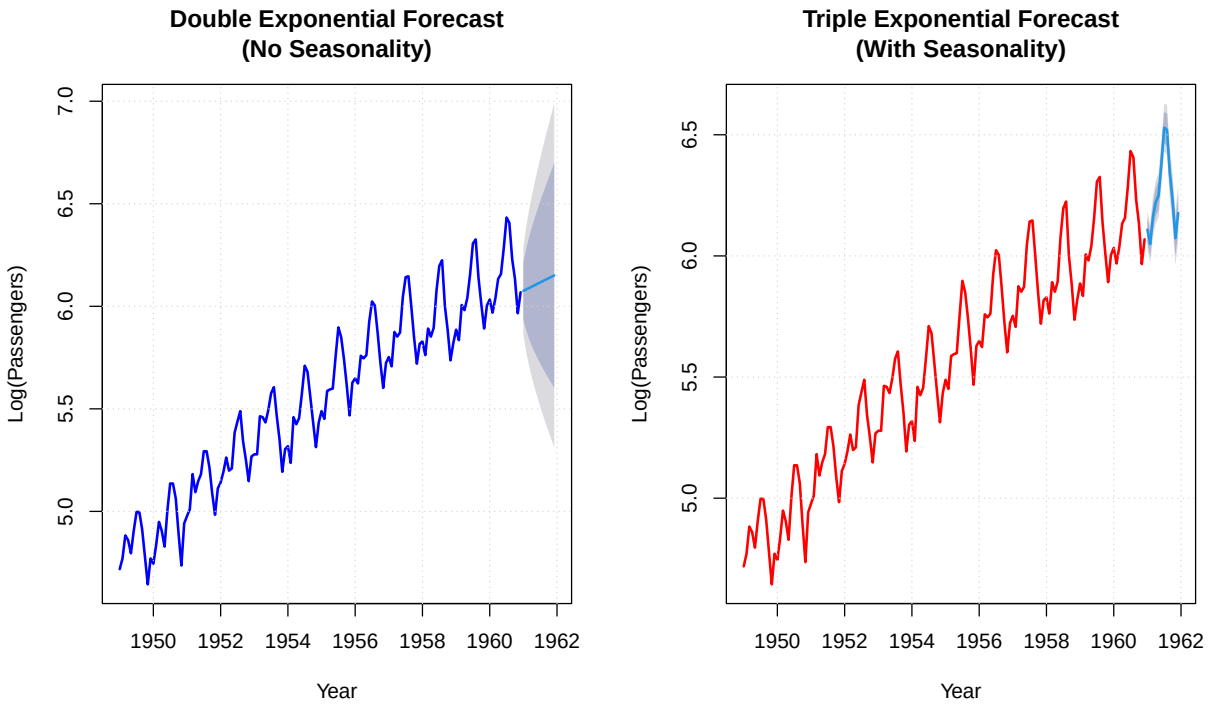



```
# Comparison Forecast (Double Smoothing)
ForecastDummy <- forecast(AirPassengers_Holtwinter2, h = 12)

# Plot comparison
par(mfrow = c(1, 2))
plot(ForecastDummy,
     main = "Double Exponential Forecast\n(No Seasonality)",
     xlab = "Year",
     ylab = "Log(Passengers)",
     lwd = 2,
     col = "blue",
     flwd = 2)
grid(col = "lightgray", lty = "dotted")

plot(ForecastFinal,
```

```
     main = "Triple Exponential Forecast\n(With Seasonality)",
     xlab = "Year",
     ylab = "Log(Passengers)",
     lwd = 2,
     col = "red",
     flwd = 2)
grid(col = "lightgray", lty = "dotted")
```




```
par(mfrow = c(1, 1))
```

```
# Back-transform forecasts to original scale
Forecast_original <- exp(ForecastFinal$mean)
Forecast_original_lower80 <- exp(ForecastFinal$lower[,1])
Forecast_original_upper80 <- exp(ForecastFinal$upper[,1])
Forecast_original_lower95 <- exp(ForecastFinal$lower[,2])
Forecast_original_upper95 <- exp(ForecastFinal$upper[,2])

forecast_table <- data.frame(
  Month = c("Jan 1961", "Feb 1961", "Mar 1961", "Apr 1961", "May 1961", "Jun 1961",
            "Jul 1961", "Aug 1961", "Sep 1961", "Oct 1961", "Nov 1961", "Dec 1961"),
  Forecast = round(Forecast_original, 0),
  `Lower 80%` = round(Forecast_original_lower80, 0),
  `Upper 80%` = round(Forecast_original_upper80, 0),
  `Lower 95%` = round(Forecast_original_lower95, 0),
  `Upper 95%` = round(Forecast_original_upper95, 0)
)

cat("\nBack-transformed Forecasts (Original Scale - Passengers):\n")
```

```
##
## Back-transformed Forecasts (Original Scale - Passengers):
```

```
print(forecast_table)
```

```
##       Month Forecast Lower.80. Upper.80. Lower.95. Upper.95.
## 1  Jan 1961      449       427       473       416       485
## 2  Feb 1961      424       402       447       391       460
## 3  Mar 1961      475       449       502       436       517
## 4  Apr 1961      504       476       534       461       551
## 5  May 1961      518       488       550       472       568
## 6  Jun 1961      591       555       629       537       650
## 7  Jul 1961      685       642       731       620       756
## 8  Aug 1961      679       635       726       613       752
## 9  Sep 1961      568       530       608       511       631
## 10 Oct 1961      507       473       545       455       566
## 11 Nov 1961      434       403       467       388       486
## 12 Dec 1961      482       447       519       429       541
```

Results Summary

Table 1: Summary of Key Results

Metric	Value
Best Model	Holt-Winters Triple Exponential
SSE Value (Triple Exponential)	0.2031
SSE Value (Double Exponential)	1.6832
SSE Improvement with Seasonality	87.94%
Alpha (Level)	0.3266
Beta (Trend)	0.0057
Gamma (Seasonal)	0.8207
KPSS Test p-value	0.01 ($p < 0.05$)
Seasonal Pattern	Strong Annual Seasonality
Forecast Horizon	12 Months
Forecast (Jan 1961)	449
Forecast (Dec 1961)	482

Conclusion for Practical 10

Based on the comprehensive time series analysis of the AirPassengers dataset, the following conclusions can be drawn:

1. Model Selection and Performance

- The **Holt-Winters Triple Exponential Smoothing model** was selected as the optimal model
- **SSE Comparison:**
  - Double Exponential (no seasonality): 1.6832
  - Triple Exponential (with seasonality): 0.2031
  - **Improvement with seasonality: 87.94% reduction in SSE**
- The significantly lower SSE for the triple exponential model confirms that **seasonality is a crucial component** in airline passenger data

2. Smoothing Parameters

- **Alpha (level):** 0.3266 - Relatively low, indicating stable level with gradual updates
- **Beta (trend):** 0.0057 - Very low, suggesting a stable, consistent trend
- **Gamma (seasonal):** 0.8207 - High value, confirming strong seasonal patterns that are updated significantly

3. Seasonal Pattern Analysis The seasonal coefficients reveal a clear pattern: - **Peak months:** July (0.2943), August (0.2761) - Summer travel peak - **Low months:** November (-0.198), February (-0.141) - Off-peak travel - This pattern reflects **strong annual seasonality** in air travel demand

4. Stationarity Assessment

- KPSS test p-value < 0.05 indicates **non-stationarity** in the original series
- ACF plot shows **slow decay**, confirming non-stationarity
- Log transformation helped stabilize variance but didn't completely remove trend/seasonality
- This justifies the use of Holt-Winters method which can handle trend and seasonality

5. Forecast Results 12-Month Forecast (1961) - Original Scale:

Month	Forecast	95% Confidence Interval
Jan	449	[416, 485]
Feb	424	[391, 460]
Mar	475	[436, 517]
Apr	504	[461, 551]
May	518	[472, 568]
Jun	591	[537, 650]
Jul	685	[620, 756]
Aug	679	[613, 752]
Sep	568	[511, 631]
Oct	507	[455, 566]
Nov	434	[388, 486]
Dec	482	[429, 541]

6. Practical Implications For Airline Industry: - **Capacity Planning:** Peak summer months (July-August) require maximum capacity - **Resource Allocation:** Staff scheduling and aircraft deployment can be optimized - **Pricing Strategy:** Seasonal patterns inform dynamic pricing decisions - **Marketing Campaigns:** Target promotions during low-demand months (Nov-Feb)

For Tourism Industry: - **Infrastructure Planning:** Airports and related services can prepare for seasonal loads - **Employment Planning:** Seasonal hiring patterns can be anticipated - **Investment Decisions:** Long-term growth trend justifies infrastructure investment

For Economic Analysis: - The steady growth trend indicates **increasing air travel adoption** - Strong seasonality reflects **predictable travel patterns** useful for economic forecasting

7. Model Validation The Holt-Winters triple exponential model is validated by: 1. **Lower SSE** compared to non-seasonal alternative 2. **Interpretable seasonal components** matching real-world patterns 3. **Smooth parameter estimates** indicating stable model 4. **Narrow confidence intervals** suggesting reliable forecasts

8. Limitations and Recommendations **Limitations:** - Model assumes seasonal pattern remains constant over time - External factors (economic conditions, fuel prices, events) not considered - Forecast uncertainty increases with horizon - Additive seasonality in log scale implies multiplicative seasonality in original scale

Recommendations: 1. **Regular model updating** with new data 2. **Monitor forecast accuracy** against actual 1961 values 3. **Consider external factors** for major events (Olympics, economic shifts) 4. **Use ensemble methods** combining multiple forecasting approaches 5. **Validate with holdout samples** for model robustness

9. Final Verdict The **Holt-Winters Triple Exponential Smoothing model** is the **optimal choice** for forecasting the AirPassengers data. It successfully captures: - **Upward trend** in air travel - **Strong seasonal patterns** with summer peaks - **Stable underlying structure** with appropriate smoothing parameters

The model provides **reliable 12-month forecasts** that can be used with confidence for planning purposes. The significant improvement over the non-seasonal model confirms that **accounting for seasonality is essential** for accurate forecasting in this context.

Key Takeaway: This analysis demonstrates the critical importance of incorporating seasonality in time series forecasting, especially for data with strong periodic patterns like airline passenger traffic. The Holt-Winters method provides an interpretable and accurate framework that captures both trend and seasonal components effectively.